

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408868
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Mirov; Russell et al.

Systems and methods for peel-off detection for low-power electronics

Abstract

Systems and methods for peel-off detection for lower power electronics are disclosed. In one embodiment, a device includes: a housing; sensor electronics disposed within the housing; an electrode positioned to receive a charge and output a control signal upon removal of a material positioned proximate to the electrode; an electronic switch configured to control an electrical connection between a power supply and the sensor electronics; and an amplifier electrically coupled to the electrode and configured to receive the control signal from the electrode and output an amplified signal to change a state of the electronic switch based on the control signal.

Inventors: Mirov; Russell (Los Altos, CA), Krasnow; Benjamin (Redwood City, CA), Schleicher; Brett (San Francisco, CA)

Applicant: Verily Life Sciences LLC (South San Francisco, CA)

Family ID: 96950352

Assignee: Verily Life Sciences LLC (Dallas, TX)

Appl. No.: 16/791187

Filed: February 14, 2020

Related U.S. Application Data

us-provisional-application US 62806001 20190215

Publication Classification

Int. Cl.: A61B5/00 (20060101); A61B5/145 (20060101); A61B5/1477 (20060101); G01D11/24 (20060101)

U.S. Cl.:

CPC **A61B5/68335** (20170801); **A61B5/0002** (20130101); **A61B5/14532** (20130101);
 A61B5/1477 (20130101); **A61B5/6833** (20130101); **G01D11/245** (20130101);
 A61B2560/0209 (20130101)

Field of Classification Search

CPC: A61B (5/68335); A61B (5/0002); A61B (5/14532); A61B (5/1477); A61B (5/6833);
 A61B (2560/0209); G01D (11/245); G06F (1/3206)

USPC: 600/347

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9431354	12/2015	Cabral, Jr. et al.	N/A	N/A
2002/0049389	12/2001	Abreu	600/318	G02C 7/04
2009/0076336	12/2008	Mazar	600/300	A61B 5/6843
2010/0286607	12/2009	Saltzstein	604/93.01	A61B 5/6848
2016/0354072	12/2015	Swift	N/A	A61B 90/30
2018/0263538	12/2017	Heikenfeld	N/A	A61B 5/14521
2018/0289973	12/2017	Carney	N/A	A61N 1/3962
2020/0008675	12/2019	Heikkilä	N/A	A61B 5/02438
2020/0253525	12/2019	Zhang	N/A	H05K 1/11

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101706499	12/2016	KR	N/A

OTHER PUBLICATIONS

Induced. 2019. In Merriam-Webster.com. Retrived Feb. 13, 2019, from <https://www.merriam-webster.com/dictionary/induce> (Year: 2019). cited by examiner

Fiveable. “Induced Charge—Principles of Physics II.” Edited by Becky Bahr, Fiveable, 2024, <https://library.fiveable.me/key-terms/principles-physics-ii/induced-charge>. Accessed May 14, 2025. (Year: 2024). cited by examiner

Wizeprep. “1.1.2 Conductors, Insulators, and Induced Charge”, 2025, <https://www.wizeprep.com/online-courses/20116/chapter/1/core/1/2>. Accessed May 14, 2025. (Year: 2025). cited by examiner

A. Pedersen, G. C. Crichton and I. W. McAllister, “The functional relation between partial discharges and induced charge,” in IEEE Transactions on Dielectrics and Electrical Insulation, vol. 2, No. 4, pp. 535-543, Aug. 1995, doi: 10.1109/94.407019. (Year: 1995). cited by examiner

Primary Examiner: Sims; Jason M

Assistant Examiner: Kretzer; Kyle W.

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 62/806,001, titled “Systems And Methods For Peel-Off Detection For Low-Power Electronics,” filed Feb. 15, 2019, the entirety of which is hereby incorporated by reference.

BACKGROUND

(1) Single use wearable sensors are becoming more popular. For many such sensors the period of time the sensor is worn by the user is a relatively small percentage of the lifetime of the sensor. For example, these sensors may include active components, e.g., a processor, memory, sensor, etc., which are powered by an onboard battery. However, one or more of these components may consume energy when the device is technically “off.” For example, in some devices the processor must remain in a “low power” mode to detect whether the user has pressed the “on” button. However, storage and shipment prior to use may take many months. Thus, a significant portion of the total battery capacity may be consumed during storage prior to activation by the end user.

SUMMARY

(2) In one embodiment, a system of the present disclosure may comprise: a device comprising: a housing; sensor electronics disposed within the housing; an electrode positioned to receive a charge and output a control signal upon removal of a material positioned proximate to the electrode; an electronic switch configured to control an electrical connection between a power supply and the sensor electronics; and an amplifier electrically coupled to the electrode and configured to receive the control signal from the electrode and output an amplified signal to change a state of the electronic switch based on the control signal.

(3) Another embodiment of the present disclosure may comprise: an electronic device comprising: an electrode configured to output a control signal upon removal of a material; an electronic switch configured to control an electrical connection between a power supply and one or more active components; and an amplifier configured to receive the control signal from the electrode and output a signal to change a state of the electronic switch based on the control signal.

(4) In one embodiment, a method of the present disclosure may comprise a method for activating an electronic device comprising: receiving, by an electrode of the electronic device, a charge in response to removal of a material from a housing of the electronic device; amplifying the charge; and closing a switch based on the amplified charge to provide power to one or more active components.

(5) These illustrative examples are mentioned not to limit or define the scope of this disclosure, but rather to provide examples to aid understanding thereof. Illustrative examples are discussed in the Detailed Description, which provides further description. Advantages offered by various examples may be further understood by examining this specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure is set forth more particularly in the remainder of the specification. The specification makes reference to the following appended figures.

(2) FIG. 1A shows an illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(3) FIG. 1B shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(4) FIG. 1C shows another illustrative system for peel-off detection for low-power electronics

according to one embodiment of the present disclosure;

(5) FIG. 1D shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(6) FIG. 2 shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(7) FIG. 3 shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(8) FIG. 4 shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(9) FIG. 5 shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure;

(10) FIG. 6 shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure; and

(11) FIG. 7 shows a flow chart of an illustrative method for peel-off detection for low-power electronics according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) Reference will now be made in detail to various and alternative illustrative embodiments and to the accompanying drawings. Each example is provided by way of explanation, and not as a limitation. It will be apparent to those skilled in the art that modifications and variations can be made. For instance, features illustrated or described as part of one embodiment may be used in another embodiment to yield a still further embodiment. Thus, it is intended that this disclosure include modifications and variations as come within the scope of the appended claims and their equivalents.

Illustrative Example of Peel-Off Detection for Low-Power Electronics

(13) One illustrative embodiment of the present disclosure includes a single use analyte sensor. For example, one such device may include a single use glucose sensing patch. Such a patch may be worn by the user for a predetermined time to measure the user's glucose levels during that time and then discarded and replaced by a new patch. The illustrative glucose sensing patch includes active components, e.g., a processor, memory, sensor, etc., which are powered by an onboard battery as well as circuitry to prevent these active components from consuming energy until the device is in use. Thus, the illustrative glucose sensing patch can be stored for a longer period of time and/or be manufactured with a smaller or less expensive battery than a conventional patch.

(14) The illustrative glucose sensing patch includes an electrode covered by a passive material (e.g., a mylar tape). When the passive material is removed a separation charge or induced charge generates a voltage across the electrode. This voltage is output to an amplifier circuit, e.g., a single transistor amplifier (e.g., an npn transistor configured as a common emitter amplifier). The amplifier circuit receives the voltage from the electrode, amplifies this voltage, and outputs the amplified voltage to an electronic switch (e.g., a transistor (e.g., a BJT or MOSFET), a relay, an electronically controlled switch, or other switching circuit) to control the flow of current from a power supply (e.g., a battery) to the active components (e.g., processor, memory, sensors, network interfaces, etc.). Alternatively, in some embodiments the output of the amplifier may be used to wake a processor from a deep-sleep state, such as by disabling a RESET line on the processor or activating a clock.

(15) In the illustrative glucose sensing patch, once the switch is closed and the active components are powered, the sensor activates and measures glucose (or some other analyte). The processor records these measurements in a memory. After a period of time the illustrative patch may transmit the stored data to a remote device (e.g., a mobile device) via a network connection (e.g., a Bluetooth, WiFi, or NFC network connection).

(16) In some embodiments the passive material may cover certain components to ensure that the user removes the passive material and thus activates the glucose sensing patch. For example, in

some embodiments, the passive material may be positioned to cover parts of components that need to be positioned to activate the glucose sensing patch. In such an embodiment, the passive material may cover sensor components in addition to the electrode (e.g., a sensor probe or wire). In other embodiments, the passive material may cover information, e.g., network connection information, thus providing a requirement that the user remove the passive material. In still other embodiments the passive material may couple the glucose sensing patch to its packaging. Thus, removing the passive material, and thereby activating the glucose sensing patch, is a necessary step in removing the glucose sensing patch from its packaging.

(17) The solution described above enables the illustrative glucose sensing patch to have a longer battery life because active components in the glucose sensing patch (e.g., the processor, memory, etc.) do not draw power from the battery until activated by removing the tape. This may enable a relatively smaller, lighter, and cheaper battery to be used than would be required without the embodiments described herein. Further, tape on an electrode is desirable because it enables the illustrative glucose sensing patch to be hermetically sealed, which may not be possible with another mechanical solution. For example, both a switch or removable tape that breaks the connection to a battery require moving components that may prevent the device from being sealed, or may make it more difficult to hermetically seal the device.

(18) Another illustrative embodiment according to the present disclosure may comprise a device for delivery of therapeutics, e.g., a wearable device with electronics for automatically delivering controlled amounts of a therapeutic, for example wearable pumps for insulin delivery. Another embodiment may be a hand-held drug delivery device with electronics for capturing and tracking drug delivery parameters such as the time that a drug was delivered or the amount of dose selected or delivered. Like a wearable sensor, such a device includes powered electronics and may have a long shelf-life prior to use. An illustrative device for delivering therapeutics may comprise an electrode covered by a passive material (e.g., a mylar tape). When the passive material is removed a separation charge or induced charge generates a voltage across the electrode. This voltage is output to an amplifier circuit, e.g., a single transistor amplifier (e.g., an npn transistor configured as a common emitter amplifier). The amplifier circuit receives the voltage from the electrode, amplifies this voltage, and outputs the amplified voltage to an electronic switch (e.g., a transistor (e.g., a BJT or MOSFET), a relay, an electronically controlled switch, or other switching circuit) to control the flow of current from a power supply (e.g., a battery) to the active components (e.g., processor, memory, sensors, network interfaces, etc.) of the device. Alternatively, in some embodiments the output of the amplifier may be used to wake a processor from a deep-sleep state, such as by disabling a RESET line on the processor or activating a clock. Such an embodiment enables longer life for the device because active components either draw no power or substantially less power prior to removal of the passive material.

(19) In yet another embodiment, a system according to the present disclosure may comprise a smart pill bottle comprising sensor electronics to sense changes in electrical, optical, acoustic or magnetic properties that occur as the lid is rotated with respect to the bottle or as the pill cap is removed from the bottle. This information may be used to measure a user's adherence to a treatment regimen and/or to prevent unauthorized access to substances stored in the pill bottle. As described above, the active components may be in a deep sleep state or unpowered until a passive material is removed from an electrode on the exterior of the device, thus preventing the battery that powers these active components from being degraded during shipment and storage prior to use.

(20) These illustrative examples are mentioned not to limit or define the scope of this disclosure, but rather to provide examples to aid understanding thereof. Illustrative examples are discussed in the Detailed Description, which provides further description. Advantages offered by various examples may be further understood by examining this specification.

Illustrative Systems for Peel-Off Detection for Low-Power Electronics

(21) FIG. 1A shows an illustrative system **100** for peel-off detection for low-power electronics

according to one embodiment of the present disclosure. As shown in FIG. 1A, the system **100** is an electronic device **114** mounted on a wearer's skin (e.g., at a location on which a sensor of electronic device **114** will be used to take measurements). System **100** further shows a passive material **116** that has been removed from an electrode on electronic device **114** to activate the electronic device **114** as is described in the following paragraphs. In some embodiments, this passive material **116** may comprise a carrier sheet that the device is removed from to expose an adhesive surface of the device that is then affixed to the user's skin.

(22) Turning now to FIG. 1B, which shows another embodiment of a system according to the present disclosure. As shown in FIG. 1B, Continuous Glucose Monitor (CGM) system **120** includes a material **126**, CGM applicator **122**, and the CGM **124**. In order to activate the CGM **124** the user removes material **126**. This sends a charge to an amplifier which applies power to active components in the CGM **124**, e.g., by closing a switch or waking a processor from a deep sleep state.

(23) Turning now to FIG. 1C, which shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure. FIG. 1C shows a delivery system **160** including a drug cartridge **162**, containing an amount of a therapeutic, an applicator **164** (e.g., a needle or cannula), and material **166**. The delivery system **160** is configured to deliver the therapeutic (e.g., insulin, or some other drug or therapeutic) to a wearer via applicator **164** (e.g., a needle). The delivery system **160** may comprise electronics that deliver the therapeutic **162** at programmed times and/or quantities. In order to activate the delivery system **160**, the user removes material **166**. This sends a charge to an amplifier which applies power to active components in the delivery system **160**, e.g., by closing a switch or waking a processor from a deep sleep state.

(24) Turning now to FIG. 1D, which shows another illustrative system for peel-off detection for low-power electronics according to one embodiment of the present disclosure. FIG. 1D shows an injection pen system **170** includes injection pen **101** comprising a drug cartridge **111**, which may be in communication with a remote device, e.g., a smartphone, via wireless communication (e.g., Wi-Fi, Bluetooth, BLE, or cellular).

(25) In the embodiment shown in FIG. 1D, drug cartridge **111** includes cartridge body **113**, and plunger head **115**. In the depicted embodiment, plunger head **115** starts near the rear of drug cartridge **111** and is pushed forward in drug cartridge **111** (with a dosage injection mechanism disposed in injection pen **101**). This forces medication/fluid out of the narrow end of drug cartridge **111** when a user chooses to dispense a fluid.

(26) Injection pen **101** is a hand-held device and includes needle **103**, body/housing **107** (including a dosage injection mechanism to push in plunger head **115** and extract fluid from drug cartridge **111**), and drug delivery control wheel **109** (twist wheel **109** to 'click' select the dosage), and pen button **150** (push button **109** to dispense the selected quantity of the fluid from cartridge **111**). Housing **107** is configured to accept cartridge **111**, which may be disposed in an insert which screws/snaps onto the bulk of housing **107**. Injection pen **101** includes a dosage injection mechanism positioned in the housing **107** to produce a rotational motion and force the fluid out of the cartridge when the drug injection pen **101** dispenses the fluid. A dosage measurement system is also disposed in the pen (e.g., in button **150**, stopper **115** or elsewhere in pen body **107**) to receive a rotational motion from the dosage injection mechanism. The dosage measurement system may measure a strain induced in a portion of the dosage measurement system by the rotational motion, and the dosage measurement system outputs a signal indicative of the strain when the drug injection pen **101** dispenses the fluid.

(27) A controller is also disposed in drug injection pen **101**, and is coupled to the dosage measurement system. This controller is configured to track data associated with the dispensation of the fluid. For example, the controller may track the time, quantity, and frequency that a fluid is dispensed. This information may be stored to measure a user's adherence to a prescribed treatment

regimen. Power to active components, such as the controller, is controlled to prevent consumption of power during shipping and storage, thus increasing overall battery life. To activate the injection pen system **170**, the user removes material **176**. This sends a charge to an amplifier which applies power to active components in the injection pen system, e.g., by closing a switch or waking a processor from a deep sleep state.

(28) Turning now to FIG. **2**, which shows an illustrative system **200** for peel-off detection for low-power electronics according to one embodiment. The system **200** includes wearable device **201**, which may include a wearable analyte sensor, e.g., a wearable glucose sensor, hydration sensor, oxygen sensor, or other type of sensor, e.g., a temperature sensor, motion sensor, blood pressure sensor, heart rate sensor, metabolic sensor, or any other type of sensor known in the art. In one embodiment, wearable device **201** includes a wearable patch comprising each of the components shown in FIG. **2**.

(29) The wearable device **201** includes a plurality of components that may have an active mode, e.g., a mode in which the device must receive power to operate. These active components include, e.g., processor(s) **202**, memory **204**, antenna **206**, and sensor **208**. In some embodiments, these components are not all active components, e.g., in some embodiments, sensor **208** and memory **204** may have both active and passive modes. Alternatively, in some embodiments, one or more of these components may include a passive component, e.g., sensor **208** may be a component that does not have an active mode.

(30) Turning to processor **202**, processor **202** comprises one or more of a microprocessor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), field programmable gate arrays (FPGAs), and state machines. Such processors may further include programmable electronic devices such as PLCs, programmable interrupt controllers (PICs), programmable logic devices (PLDs), programmable read-only memories (PROMs), electronically programmable read-only memories (EPROMs or EEPROMs), or other similar devices.

(31) Processor **202** is in communication with memory **204**. Memory **204** can include any suitable tangible (and non-transitory) computer-readable medium such as RAM, ROM, EEPROM, or the like, and may embody program components that configure operation of the processor **202** and/or other components of device **201**. Memory **204** may further have storage for processor **202** to store data, e.g., data received from a network interface or sensor **208**. Memory **204** may be a static memory that can store data while in an unpowered state.

(32) Antenna **206** represents one or more of any components that facilitate a network connection. Examples include, but are not limited to, wired interfaces such as Ethernet, USB, IEEE 1394, and/or wireless interfaces such as IEEE 802.11, Bluetooth, or radio interfaces for accessing cellular telephone networks (e.g., transceiver/antenna for accessing a CDMA, GSM, UMTS, or other mobile communications network(s)). Thus, antenna **206** enables processor **202** to transmit or receive data from a remote device.

(33) Sensor **208** comprises a sensor that measures information about a user, e.g., an analyte sensor, e.g., a wearable glucose sensor, hydration sensor, oxygen sensor, or other type of sensor, e.g., a temperature sensor, motion sensor, blood pressure sensor, heart rate sensor, metabolic sensor, or any other type of sensor known in the art. In some embodiments, sensor **208** may be an active sensor, a passive sensor, or a sensor with both active and passive modes.

(34) As shown in FIG. **2**, wearable device **201** further includes a gated power supply **250**. The gated power supply **250** includes a power supply **210**, a switch **212**, an electrode **214**, material **216**, and amplifier **218**.

(35) Power supply **210** is a source of power, e.g., an onboard power source such as a battery, e.g., a rechargeable or non-rechargeable battery, for example, an alkaline battery, aluminum-air battery, aluminum-ion battery, dry cell, lithium battery, magnesium battery, mercury battery, nickel oxyhydroxide battery, lead-acid battery, lithium air battery, lithium ion battery, magnesium ion battery, nickel cadmium battery, sodium ion battery, or any other type of battery known in the art.

(36) Switch **212** is a switch that interrupts the flow of current from power supply **212** to one or more of the active components of wearable device **201**. For example, in some embodiments, switch **212** may be a transistor (e.g., a BJT or MOSFET), a relay, an electronically controlled switch, an op-amp circuit, or other switching circuit, e.g., a solid-state or non-solid state circuit that interrupts the flow of current from a power source to a load. Further, after switch **212** is closed it may automatically reopen after a predetermined period of time, e.g., one week or two weeks, or it may stay permanently closed. Further, in some embodiments, switch **212** may be controlled by processor(s) **202** to be opened after the predetermined period of time, thus cutting the flow of current from power supply **210** to the active components.

(37) Turning now to electrode **214**, which is a conductive electrical material. In some embodiments, electrode **214** may be a conductor (e.g., copper, aluminum, silver etc.) etched on the inside or outside of a housing associated with device **201**. For example, in some embodiments, the electrode **214** may include a copper pattern from a printed circuit board (PCB). The electrode **214** is covered in whole or in part by material **216**, which may be a passive material such as tape or mylar. In some embodiments mylar may provide electrostatic shielding, such that there is no false triggering of a charge onto electrode **214** until the mylar is removed. Removing material **216** generates a charge on electrode **214**. In some embodiments, electrode **214** may have a rectangular shape that is approximately 4×9 mm. In some embodiments, electrode **214** may be positioned on the interior of a housing of device **201**. In such an embodiment, the material **216** may be placed on the exterior of the housing at a location above the location on which the electrode is positioned.

(38) Electrode **214** is further coupled to amplifier **218**. When a charge is generated across electrode **214**, amplifier **218** amplifies that charge into a control signal that changes the state of switch **212**. In some embodiments, amplifier **218** may be a simple amplifier circuit, e.g., an NPN transistor amplifier with minimal components (e.g., a transistor, resistor, and capacitor). In other embodiments, amplifier **218** may be an op-amp circuit or a transistor circuit (e.g., BJT, FET, MOSFET, or HBT circuit) that amplifies an input signal.

(39) In some embodiments material **216** may be tape that prevents a housing of an electronic device from being opened. In such an embodiment, the memory **204** may store data associated with when material **216** was removed. This information may be used to determine if a user has voided the warranty on an electronic device, e.g., by removing the tape to open the case of the device.

(40) In some embodiments, device **201** may have a plastic case that is hermetically sealed. In some embodiments, device **201** may be an electronic device that is manufactured and shipped with switch **212** open, such that very little if any power is consumed until a user receives the electronic device and removes material **216**, thus closing switch **212**.

(41) Turning now to FIG. 3, FIG. 3 shows another illustrative system **300** for peel-off detection for low-power electronics according to one embodiment of the present disclosure. As shown in FIG. 3, the system **300** includes an electrode **314** covered by material **316** (which are similar to electrode **314** and material **316** described with regard to FIG. 1A), amplifier **318**, and controller **315**.

(42) In the embodiment shown in FIG. 3, when a user removes material **316** from electrode **314**, a charge is generated and transferred to the electrode **314**. This charge is then amplified by amplifier **318**. As shown in FIG. 3, the amplifier **318** is a single transistor amplifier circuit. The transistor **304** may comprise a 2N3904 NPN Transistor. The collector may include a 10 MΩ resistor **302** coupled to a voltage (e.g., a voltage provided by power supply **210** discussed above with regard to FIG. 2). The transistor **204** is DC coupled by capacitor **206**, which may be a 220 picofarad capacitor.

(43) In the embodiment shown in FIG. 3, the amplifier **318** may amplify the input signal generated when the material **316** is removed from electrode **314**). This amplified signal is output by amplifier **318** as a control signal to controller **350**. Controller **350** may comprise an electronic switch (e.g., switch **212** described above with regard to FIG. 2). In such an embodiment, the control signal closes the electronic switch and thus provide power to active components (e.g., a processor, a memory, a sensor, etc.). Further, in some embodiments, controller **350** may include a processor. In

such an embodiment, the processor may be in a deep-sleep state be woken out of the deep-sleep state by the control signal from amplifier **318**.

(44) Turning now to FIG. **4**, FIG. **4** shows another illustrative system **400** for peel-off detection for low-power electronics according to one embodiment of the present disclosure. As shown in FIG. **4**, the system **400** includes housing **402** electrode **414**, material **416**, and sensor **408**. Housing **402** is a housing for an electronic device, e.g., a hermetically sealed housing for a CGM, as described above. The electrode **414** and material **416** are similar to electrode **114** and material **116** described above. As is shown in FIG. **4**, the material **416** is positioned such that it must be removed prior to using an associated electronic device. In the embodiment shown in FIG. **4**, the material **416** covers part of a sensor **408**, e.g., a probe, port, electrode, or sensor wire associated with sensor **408**. For example the sensor **408** may be an electrode for an electrocardiogram (“ECG”) sensor. Thus, to place the sensor in position the user must remove the material **416** and thereby activate the associated electronic device, as described above.

(45) Turning now to FIG. **5**, FIG. **5** shows another illustrative system **500** for peel-off detection for low-power electronics according to one embodiment of the present disclosure. As shown in FIG. **5**, the system **500** includes housing **502** electrode **514**, material **516**, and information **520**. Housing **502** is a housing for an electronic device, e.g., a hermetically sealed housing for a CGM, as described above. The electrode **514** and material **516** are similar to electrode **114** and material **116** described above. As is shown in FIG. **5**, the material **516** is positioned such that it must be removed prior to using an associated electronic device. In the embodiment shown in FIG. **5**, the material **516** covers information **520**. Information **520** may be information needed to activate an associated device, e.g., instructions, login information, passkey information, etc. In other embodiments, the information **520** is printed on the underside of material **516**. Thus, in order to use an associated electronic device the user must remove the material **516** and thereby activate the associated electronic device, as described above.

(46) Turning now to FIG. **6**, FIG. **6** shows another illustrative system **600** for peel-off detection for low-power electronics according to one embodiment of the present disclosure. As shown in FIG. **6**, the system **600** includes housing **604** packaging **602**, electrode **614**, and material **616**. Housing **604** is a housing for an electronic device, e.g., a hermetically sealed housing for a CGM, as described above. The electrode **614** and material **616** are similar to electrode **114** and material **116** described above. As is shown in FIG. **6**, the material **616** is positioned such that it must be removed prior to using an associated electronic device. In the embodiment shown in FIG. **6**, the material **616** couples housing **604** to packaging **602**. Thus, removing housing **604** from packaging **602** requires also removing material **616** from electrode **614** and thereby activate the associated electronic device, as described above.

Illustrative Methods for Peel-Off Detection for Low-Power Electronics

(47) FIG. **7** is a flow chart for a method **700** of peel-off detection for low-power electronics. Alternatively, in some embodiments, one or more of the steps shown in FIG. **7** may be skipped, or additional steps not shown in FIG. **7** may be performed. The steps below are described with reference to components described above with regard to system **200** shown in FIG. **2**.

(48) The method **700** begins at step **702**, when a material **216** is moved from an electrode **214**. Prior to removal of the material **216**, the electrode **214** is covered in whole or in part by material **216**, which may be a passive material such as tape or mylar. Removing material **216** generates a charge on electrode **214**. In some embodiments, the material **216** may further cover part of sensor **208** (e.g., a sensor wire, port, or probe), thus the material **216** must be removed to use the sensor. In other embodiments, the material **216** may cover information (e.g., instructions, login information, passkey information, etc.). Further, in some embodiments, this information may be printed on the underside of material **216**. In either case, the user must remove the material to read the information. In still other embodiments, the material **216** may couple the device **201** to packaging (e.g., for shipping). In such an embodiment, the user must remove the material from the electrode **214** to

remove the device **201** from its packaging.

(49) The at step **704** amplifier **218** receives a control signal from electrode **214**. As described above, removing a material **216** from electrode **214** generates a voltage on electrode **214**.

(50) Next at step **706** the amplifier **118** amplifies the control signal. Amplifier **218** is an amplifier circuit that amplifies the voltage on the electrode **214** and apply that amplified voltage as a control signal to switch **212**.

(51) Then at step **708** the switch **212** applies power to one or more active components based on the control signal. In one embodiment, the control signal directly triggers the switch **212**. In such an embodiment, switch **212** is electrically coupled to amplifier **218** in a configuration such that current from amplifier **218** closes switch **212** and thus provides current to one or more active components on wearable device **201**, e.g., processor **202**, memory **204**, and sensor **208**. In some embodiments, the control signal may indirectly control switch **212**, for example, by triggering analog or digital circuitry to apply current to switch **212**. Further, in some embodiments, switch **212** may be reopened, for example, automatically after a predetermined period of time, or after receipt of a control signal from processor **202**.

(52) Then at step **710** processor **202** receives signals from sensor **208**. The sensor **208** measures information about a user, e.g., sensor **208** may be an analyte sensor, e.g., a wearable glucose sensor, hydration sensor, oxygen sensor, or other type of sensor, e.g., a temperature sensor, motion sensor, blood pressure sensor, heart rate sensor, metabolic sensor, or any other type of sensor known in the art. In some embodiments, sensor **208** may be an active sensor, a passive sensor, or a sensor with both active and passive modes.

(53) Next, at step **712** the processor **202** stores data received from the sensor **208** in memory **204**. Memory **204** may be a static memory, e.g., flash or EEPROM, to store the sensor data received from processor **202** after switch **212** is reopened and power is no longer provided to the active components in wearable device **201**.

(54) Then at step **714** the processor transmits the stored data to a remote device. In such an embodiment, the processor **202** may enter a transmit mode, in which the processor **202** transmits the data stored in memory **204** using antenna **206**.

(55) The solution described above enables the illustrative glucose sensing patch to have a longer battery life because active components in the glucose sensing patch (e.g., the processor, memory, etc.) do not draw power from the battery until activated by removing the tape. Thus, there is not energy consumed during manufacture, shipping, and storage. This may enable a relatively smaller, lighter, and cheaper battery to be used than would be required without embodiments described herein. Further, tape on an electrode is preferable because it enables the illustrative glucose sensing patch to be hermetically sealed, which is not possible with another mechanical solution. For example, both a switch or removable tape that breaks the connection to a battery require moving components that prevent the device from being sealed.

(56) The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

(57) Specific details are given in the description to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the

scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

(58) Also, configurations may be described as a process that is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Furthermore, examples of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks may be stored in a non-transitory computer-readable medium such as a storage medium. Processors may perform the described tasks.

(59) Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered. Accordingly, the above description does not bound the scope of the claims.

(60) The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.

(61) Embodiments in accordance with aspects of the present subject matter can be implemented in digital electronic circuitry, in computer hardware, firmware, software, or in combinations of the preceding. In one embodiment, a computer may comprise a processor or processors. The processor comprises or has access to a computer-readable medium, such as a random access memory (RAM) coupled to the processor. The processor executes computer-executable program instructions stored in memory, such as executing one or more computer programs including a sensor sampling routine, selection routines, and other routines to perform the methods described above.

(62) Such processors may comprise a microprocessor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), field programmable gate arrays (FPGAs), and state machines. Such processors may further comprise programmable electronic devices such as PLCs, programmable interrupt controllers (PICs), programmable logic devices (PLDs), programmable read-only memories (PROMs), electronically programmable read-only memories (EPROMs or EEPROMs), or other similar devices.

(63) Such processors may comprise, or may be in communication with, media, for example tangible computer-readable media, that may store instructions that, when executed by the processor, can cause the processor to perform the steps described herein as carried out, or assisted, by a processor. Embodiments of computer-readable media may comprise, but are not limited to, all electronic, optical, magnetic, or other storage devices capable of providing a processor, such as the processor in a web server, with computer-readable instructions. Other examples of media comprise, but are not limited to, a floppy disk, CD-ROM, magnetic disk, memory chip, ROM, RAM, ASIC, configured processor, all optical media, all magnetic tape or other magnetic media, or any other medium from which a computer processor can read. Also, various other devices may include computer-readable media, such as a router, private or public network, or other transmission device. The processor, and the processing, described may be in one or more structures, and may be dispersed through one or more structures. The processor may comprise code for carrying out one or

more of the methods (or parts of methods) described herein.

(64) Use herein of the word “or” is intended to cover inclusive and exclusive OR conditions. In other words, A or B or C includes any or all of the following alternative combinations as appropriate for a particular usage: A alone; B alone; C alone; A and B only; A and C only; B and C only; and A and B and C.

(65) While the present subject matter has been described in detail with respect to specific embodiments thereof, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing may readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, it should be understood that the present disclosure has been presented for purposes of example rather than limitation, and does not preclude inclusion of such modifications, variations and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art.

Claims

1. A device comprising: a housing; sensor electronics disposed within the housing; only one electrode positioned to receive an induced charge generated by removal of a material from the electrode and output a control signal, the induced charge being characterized by a voltage across the electrode; an electronic switch configured to control an electrical connection between a power supply and the sensor electronics; and an amplifier electrically coupled to the electrode and configured to receive the control signal from the electrode and output an amplified signal to change a state of the electronic switch based on the control signal.
2. The device of claim 1, further comprising a non-transitory memory configured to store data received from the sensor electronics.
3. The device of claim 2, wherein the non-transitory memory is configured to record that the material was removed.
4. The device of claim 2, wherein the electronic switch is further coupled between the power supply and a processor; and wherein the processor is configured to receive sensor data from the sensor electronics when the electronic switch is closed.
5. The device of claim 4, further comprising a network interface configured to transmit the sensor data stored in the non-transitory memory.
6. The device of claim 5, wherein the housing is hermetically sealed.
7. The device of claim 1, wherein the electronic switch comprises one or more of: a transistor, a relay, or an electronically controlled switch.
8. The device of claim 1, wherein, prior to removal, the material covers one or more of: part of a sensor or activation information positioned on the housing.
9. The device of claim 1, wherein the material comprises mylar.
10. An electronic device comprising: only one electrode configured to output a control signal based upon an induced charge generated by removal of a material on the electrode, the induced charge being characterized by a voltage across the electrode; an electronic switch configured to control an electrical connection between a power supply and one or more active components; and an amplifier configured to receive the control signal from the electrode and output a signal to change a state of the electronic switch based on the control signal.
11. The electronic device of claim 10, wherein the electronic switch, the amplifier, and the one or more active components are all positioned within a hermetically sealed housing.
12. The electronic device of claim 10, wherein the one or more active components comprise a memory configured to record that the material was removed.
13. The electronic device of claim 10, wherein the electronic switch comprises one or more of: a transistor, a relay, or an electronically controlled switch.
14. The electronic device of claim 10, wherein prior to removal the material covers one or more of:

part of a sensor or activation information positioned on a hermetically sealed housing.

15. The electronic device of claim 10, wherein the material comprises mylar.

16. A method for activating an electronic device comprising: receiving, by an electrode of the electronic device, an induced charge generated by removal of a material from a housing of the electronic device, the induced charge being characterized by a voltage across the electrode and wherein the electronic device comprises only one electrode; amplifying, by an amplifier, the induced charge; and closing a switch based on the amplified induced charge to provide power to one or more active components.

17. The method of claim 16, wherein the amplifier and the one or more active components are all positioned within a hermetically sealed housing.

18. The method of claim 16, wherein the one or more active components comprise a memory configured to record that the material was removed.

19. The method of claim 16, wherein prior to removal the material covers one or more of: part of a sensor or activation information positioned on the housing.

20. The method of claim 16, wherein the material comprises mylar.
